

1. CLO 1 (5 คะแนน) In the domain of integer, let $P(x, y)$ be the predicate " $x \cdot y = 12$."

Tell whether each of the following statements is True or False.

(a) $P(3, 4)$

(b) $P(3, 5)$

(c) $P(2, 6) \vee P(3, 7)$

(d) $(\forall x)(\forall y)(P(x, y) \rightarrow P(y, x))$

(e) $(\forall x)(\exists y)P(x, y)$

(a) Ans : True

(b) Ans : False

(c) Ans : False.

(d) Ans : False

(e) Ans : True

2. CLO 1 (8 คะแนน) In the domain of all books, consider the following predicates.

$H(x)$ = "x is a heavy."

$C(x)$ = "x is confusing."

Translate the following statements in predicate logic into ordinary English.

(a) (2 คะแนน) $(\forall x) (H(x) \rightarrow C(x))$

(b) (2 คะแนน) $(\exists x) (C(x) \wedge H(x))$

(c) (2 คะแนน) $(\forall x) (C(x) \vee H(x))$

(d) (2 คะแนน) $(\exists x) (H(x) \wedge \neg C(x))$

(a) Ans: "All x is heavy then x is confusing."

(b) Ans: "There has some x is confusing and heavy"

(c) Ans: "All x is heavy or confusing"

(d) Ans: "There has some x is heavy but not confusing."

3. CLO 1 (10 คะแนน) The domain of the following predicates is the set of all plants.

$P(x)$ = "x is poisonous."

$Q(x)$ = "Jeff has eaten x."

Translate the following statements into predicate logic.

(a) (3 คะแนน) Some plants are poisonous.

(b) (3.5 คะแนน) Jeff has never eaten a poisonous plant.

(c) (3.5 คะแนน) There are some non-poisonous plants that Jeff has never eaten.

(a) Ans : $\exists x [P(x)]$

(b) Ans : $\forall x [\neg P(x) \rightarrow \neg Q(x)]$

(c) Ans : $\exists x [\neg P(x) \rightarrow \neg Q(x)]$

4. CLO 1 (7 คะแนน) In the domain of all people, consider the following predicate.

$P(x, y)$ = "x needs to love y."

(a) (4 คะแนน) Write the statement "Everybody needs to love somebody." in predicate logic.

(b) (2 คะแนน) Formally negate your statement from part (a).

(c) (1 คะแนน) Write the English translation of your negated statement.

(a) Ans : $\forall x \exists y [P(x, y)]$

(b) Space : $\sim \forall x \exists y [P(x, y)] \equiv \exists x \forall y [\sim P(x, y)]$

Ans : $\exists x \forall y [\neg P(x, y)]$

(c) Ans : "There has somebody needs to love everybody"

5. CLO 1 (7 คะแนน) The domain of the following predicates is all integers greater than 1.

Consider the following statement.

For every x that is not prime, there is some prime y that divides it.

(a) (4 คะแนน) Write the statement in predicate logic.

(b) (2 คะแนน) Formally negate the statement.

(c) (1 คะแนน) Write the English translation of your negated statement.

(a) Ans : Let " x is prime " is $P(x)$
Let " x is divided by y " is $D(x,y)$.
So $\forall x \exists y [\neg P(x) \rightarrow (P(y) \wedge D(x,y))]$

(b) Ans : $\exists x \forall y [P(x) \wedge (\neg P(y) \wedge \neg D(x,y))]$

(c) Ans : For some x is prime, There all non-prime y that isn't divide it.

6. CLO 1 (20 คะแนน) Let the following predicates be given in the domain of all triangles.

$R(x)$ = "x is a right triangle."

$B(x)$ = "x has an obtuse angle."

Consider the following statements.

$$S_1 = \neg(\exists x)(R(x) \wedge B(x))$$

$$S_2 = (\forall x)(R(x) \rightarrow \neg B(x))$$

(a) (16 คะแนน) Write a proof sequence to show that $S_1 \Leftrightarrow S_2$.

Hint : proof sequence for showing the equivalence of S_1 and S_2 can be done as follows.

	Propositions	Reasons
1.	$\sim (\exists x) (R(x) \wedge B(x))$	given
2.		
3.		
	...	...
	...	...
N.	$\forall x (R(x) \rightarrow \sim B(x))$	

(b) (2 คะแนน) Write S_1 in ordinary English.

(c) (2 คะแนน) Write S_2 in ordinary English.

	Proposition	Reason
(a) Ans:	1. $\neg \exists(x) [R(x) \wedge B(x)]$	given
	2. $\forall(x) [\neg R(x) \vee \neg B(x)]$	De Morgan's Laws, 1.
	3. $\forall(x) [R(x) \rightarrow \neg B(x)]$	Equivalence Rules, 2

(b) Ans: "There is no some x, that x is right triangle and have an obtuse angle."

(c) Ans: "For all x is right triangle then the x hasn't an obtuse angle."

7. CLO 1 (23 คะแนน) Let the following predicates be given. The domain is all computer science classes.

$I(x)$ = "x is interesting."

$U(x)$ = "x is useful."

$H(x, y)$ = "x is harder than y."

$M(x, y)$ = "x has more students than y."

(a) (6 คะแนน) Write the following statements in predicate logic.

- All interesting CS classes are useful.
- There are some useful CS classes that are not interesting.
- Every interesting CS class has more students than any non-interesting CS class.

(b) (2 คะแนน) Write the following predicate logic statement in everyday English. Don't just give a word-for-word translation; your sentence should make sense.

$$(\exists x)[I(x) \wedge (\forall y)(H(x, y) \rightarrow M(y, x))]$$

(c) (8 คะแนน) Formally negate the statement from part (b). Simplify your negation so that no quantifier lies within the scope of a negation. State which derivation rules you are using.

(d) (2 คะแนน) Give a translation of your negated statement in everyday English.

(a) Ans : i. $\forall x [I(x) \rightarrow U(x)]$
ii. $\exists x [U(x) \vee \neg I(x)]$
iii. $\forall x \forall y [I(x) \wedge (M(x, y) \rightarrow \neg I(y))]$

(b) Ans : Some interesting CS class that's harder, would has more student than else.

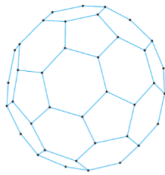
(c) Space : $\neg \exists x [I(x) \wedge (\forall y) (H(x, y) \vee M(y, x))]$

Ans : $\forall x [\neg I(x) \vee (\exists y) (H(x, y) \wedge \neg M(y, x))]$

(d) Ans : All non-interesting CS classes which harder and has less student than some else.

8. CLO 1 (20 คะแนน)

Let the domain be all faces of the following truncated icosahedron (also known as a soccer ball). Consider the following three predicates:



$P(x)$ = "x is a pentagon."

$H(x)$ = "x is a hexagon."

$B(x, y)$ = "x borders y."

Here we say that two polygons border each other if they share an edge.

We also stipulate that a polygon cannot border itself.

8.1. (10 คะแนน)

Write down the correct proposition (in predicate logic) for the statement:

"No two pentagons border each other."

8.2. (10 คะแนน)

Write negation of the proposition for the statement in 8.1.

You must show formally how you obtain the negated statement via formal logic statements.

Simply write down the negated proposition in ordinary English will not get scores.

8.1) Ans : $\forall x \forall y [(P(x) \wedge P(y)) \rightarrow \neg (B(x, y))]$

8.2) Ans : 1 $\forall x \forall y [(P(x) \wedge P(y)) \rightarrow \neg (B(x, y))]$, given

2. $\neg \forall x \forall y [(P(x) \wedge P(y)) \rightarrow \neg (B(x, y))]$, Negation 1

3 $\neg \forall x \forall y [\neg (P(x) \wedge P(y)) \vee \neg (B(x, y))]$, implication 2

4. $\exists x \exists y [(P(x) \wedge P(y)) \wedge B(x, y)]$, De Morgan's Laws 3

English : " Some two pentagon is borders than each other